

AYUSH GEHLOT

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PROFESSIONAL EXPERIENCE

The New York Stem Cell Foundation, New York, NY | Software Engineer

June 2025 – August 2025

- Coordinated with R&D team to research & prototype smart incubator system formulated to preserve human body & blood cells under optimal environmental conditions.
- Developed a C-based PID control system to regulate O₂, CO₂, N₂, and internal pressure within the incubator.
- Built a React-based touchscreen GUI on Raspberry Pi for real-time monitoring and configuration of gas and environmental parameters.

Dream Team Technologies PVT LTD., Rajasthan, India | Full Stack Developer Intern

June 2022 – March 2023

- Collaborated with developers and UX/UI designers to build a university management platform using HTML, CSS, and JavaScript for attendance tracking, assignments, and student enrollment.
- Developed responsive user interfaces using React, Angular, and Vue.js, while implementing OAuth authentication for secure access and payment handling.
- Automated UI testing using Selenium and PyTest, reducing manual testing time by 75% and improving deployment efficiency.

Defense Research & Development Organization (DRDO), Rajasthan, India | Hardware Engineering Intern

June 2021 – March 2022

- Conducted rigorous validation of VHDL-designed circuits using Model Sim, used to design & simulate digital hardware systems, assessing syntax & structure of VHDL code through its architecture & configuration, achieving 100% verification rate, leading to increased project approvals across teams.

ACADEMIC PROJECTS & EXPERIENTIAL LEARNING /HACKATHON

TrueSight — Hybrid Perceptual-Hash and CLIP-Embedding Application for Image Authentication Verification

January 2026 - May 2026

- Collaborated in a team of 5 to develop TrueSight, a prototype for detecting AI-generated, manipulated, and authentic images using perceptual hashing and neural similarity techniques.
- Built an AI-powered verification pipeline using p-Hash, CLIP-based embeddings, and cryptographic provenance concepts for image authenticity analysis.
- Researched real-world challenges in deepfake detection, adversarial image manipulation, and provenance verification inspired by modern trust & safety systems used by major technology platforms.

Smart City Dashboard — NYC Experiential Learning Lab Sponsored by Spotify, MongoDB & Kensho Technologies (1st Place)

November 2025

- Developed a responsive React-based dashboard with geospatial maps, trend graphs, and forecasting visualizations for city-wide sanitation analysis.
- Collaborated with data engineers, data scientists, and business analysts to transform statistical models into interactive insights for operational decision-making.
- Integrated MongoDB Atlas to manage 100K+ sanitation and complaint records with real-time filtering and aggregation.

AI Restaurant Chatbot for CaterPal Technologies Hackathon (Secured 1st Place)

March 2025

- Developed an AI-powered restaurant chatbot using Python (Flask), JavaScript, MongoDB, and OpenAI API to deliver real-time food recommendations based on user preferences.
- Established surplus food donation feature to connect catering services with local charities to address food waste & social impact goals while integrating feedback system to track most recommended dishes & personalize suggestions based on user preferences.

Advanced Driving Assistant System using Arduino & C++

March 2023 – June 2023

- Synchronized in team of four to develop advanced driving assistant system to support obstacle detection & vehicle navigation, integrating sensors & cameras programmed with image processing to reduce road accidents.
- Built model car with motor for system testing, creating system with Arduino sensors, LIDAR, & Integrated Radar to detect obstacles with sonic waves, integrating night-vision image sensing to boost detection accuracy & GPS APIs to direct drivers around obstacles, refining prototype to achieve approximately 70% accuracy.
- Proposed system to India's Minister of Railways, leading to integration of prototype in ministry's systems to improve railway safety.

EDUCATION

Pace University | New York, NY | Master of Science (MS) in Computer Science | **Concentration:** Software Development | **GPA:** 3.57 May 2026

MBM University | Rajasthan, India | Bachelor of Engineering (BEng) in Electronics and Computer

June 2023

RELEVANT COURSEWORK

Object Oriented Programming | Data Structures & Algorithms | Web Technology | Statistics | VLSI Design | Operating Systems | Software Testing | Database Management | Computer Engineering | Information Technology | Cloud Computing | Application Development | Agile Development | SDLC | AI |

TECHNICAL SKILLS

Programming Languages: C, C++, C#, Java, Python, JavaScript, R, VHDL | **Frontend:** HTML5, CSS3, React.js, Vue.js, Angular, Three.js | **Backend:** Node.js, Express.js, REST APIs, .NET | **Database Management:** MySQL, NoSQL, MongoDB, PostgreSQL | **Cloud & Platforms:** AWS, EC2, S3, Microsoft Azure, | **Operating Systems:** Windows, Linux, Android, Mac OS | **Software / Tools:** VS Code, Cursor, R Studio, Tableau, Jira, Git, Postman, Docker, | **Libraries:** REACT, GTK, | **Protocols:** HTTP, TCP/IP, UDP, SSH, DNS, FTP, QUIC, DHCP, SM, IMAP | **Development Methodologies:** Agile, Scrum